

AI Intelligence Ingestion Pipeline

Sources -> Discovery -> Queue -> Async Workers

Fetch / Browser -> Raw Store -> Normalize -> Freshness Gate

LLM Router -> Validation -> Entity Resolution -> PostgreSQL

PostgreSQL -> Vector Store / Graph Store -> Google Sheets

Cross-cutting: provenance, idempotency, caching, observability, quarantine

Reliability Flows

413 CONTEXT TOO LARGE

token estimate -> semantic heading/paragraph chunks -> extraction

-> deterministic merge -> final schema validation

429 RATE LIMIT

Retry-After -> bounded exponential backoff + jitter -> retry budget

-> fallback provider -> circuit breaker -> quarantine on exhaustion

Scale and Freshness

500k+ records -> partitioned queue -> stateless workers -> shared durable storage

Scale capacity by adding workers; apply source/provider limits and backpressure

source URL -> canonical URL -> idempotency key -> database uniqueness

publication date -> priority parser -> UTC normalization -> 24-hour gate

fresh records -> watermark -> validated export; stale/future/unknown -> reject or quarantine